

ARSLAN KHAN

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[arslankhan0.github.io/portfolio](https://github.com/arslankhan0/portfolio)

Research Interests

- **Autonomous Systems & State Estimation:** Sensor fusion (Visual-Inertial Navigation Systems), SLAM, and path planning in unmanned autonomous systems (UAVs/UGVs).
- **Intelligent Control Engineering:** Nonlinear dynamics, high-gain observers, multi-axis precision motor control, and closed-loop actuation strategies.
- **Robotic Perception & Embedded AI:** Embedded computer vision, automated target tracking, real-time edge processing (YOLO, OpenCV), and integrated hardware-software architectures.

Education

National University of Sciences and Technology (NUST) — Islamabad, Pakistan

College of Electrical and Mechanical Engineering (CEME) 2018 – 2022

- **Bachelor of Science in Electrical Engineering** | CGPA: 3.08 / 4.00
- **Final Year Project / Thesis:** Implementation of SLAM & Visual-Inertial State Estimation in Quadrotors
- **Major Coursework:** Object-Oriented Programming (C++, C#), Signals and Systems, Digital Signal Processing (DSP), Digital System Design, Linear Control Systems, Computer Vision, Machine Learning.

Garrison Academy Kharian Cantt — FBISE, Pakistan

Higher Secondary School Certificate (FSc Pre-Engineering) 2016 – 2018

Federal Government Boys High School Kharian Cantt — FBISE, Pakistan

Secondary School Certificate (Matric Pre-Engineering) 2014 – 2016

Research & Professional Experience

Independent Researcher & Engineering Consultant *March 2023 – Present*

Self-Employed / Independent Project Research

- **Industrial Vehicle Automation:** Architected full-scale automation and custom control electronics for a **Toyota Stacker Industrial Forklift**, integrating motor drive circuits, embedded sensor interfaces, and automated motion planning algorithms, CAN interface decoding.
- **Embedded Engineering Consulting (Upwork):** Executed 25+ complex hardware and firmware projects globally (\$8k+ top-rated earnings), specializing in STM32 microcontrollers, ROS/ROS2 node architectures, embedded Linux platforms, and custom PCB designs.

Embedded Design Engineer *June 2023 – March 2026*

Cavalier Group of Companies — Islamabad, Pakistan

- **Project Remote Weapon System (RWS / RCWS):** Spearheaded full-scale design, development, and control system architecture for a Remote Weapon System for APCs. Designed closed-loop dual-axis motor control loops for coordinated pan-tilt actuation and integrated AI-based target tracking pipelines in C/C++.
- **Project MSV:** Directed full-scale system integration for MSV payloads, central server communication protocols, surveillance camera platforms, and automated drone-docking subsystems.
- **Project Digital Directional Finder:** Engineered hardware architecture and signal processing interfaces for a full-scale UHF/VHF direction-finding product.
- **Project Payload Dropping & Parachute Recovery Systems:** Designed and prototyped hexacopter and quadrotor platforms featuring custom mortar-dropping mechanisms and parachute deployment architectures.
- **Integrated Building Management System (IBMS):** Developed COTS-based sensor telemetry and control platforms for rapid testing and hardware prototyping.

Research Intern Aug 2022 – Oct 2022

Unmanned Vehicles Research Lab (UVRL), Department of Electrical Engineering, CEME, NUST

- Gained deep technical understanding of quadrotor flight controller hardware topologies and sensor fusion modules.
- Developed, programmed, and rigorously bench-tested embedded firmware within hardware-in-the-loop (HIL) environments.
- Performed circuit debugging, component soldering, and precision hardware troubleshooting.

Peer-Reviewed Publications

1. **A. Khan**, H. Ullah, U. Faisal, N. Mazhar, and M. Y. Khan, "*Visual-Inertial State-Estimation Using Ground Station for UAV*," 2023 4th International Conference on Computing, Mathematics and Engineering Technologies (**iCoMET**), Sukkur, Pakistan, 2023, pp. 1-6, doi: 10.1109/iCoMET57998.2023.10099145.
2. A. Raza, N. Mazhar, F. M. Malik, R. Khan, **A. Khan**, and H. Ullah, "*Output Feedback Control of Two-Time-Scale Permanent-Magnet DC Motor Using High-Gain Observers*," **ICAME 2023**, Engineering Proceedings, Sep. 2023, doi: 10.3390/engproc2023045020.
3. A. Raza, F. M. Malik, N. Mazhar, R. Khan, **A. Khan**, and H. Ullah, "*High-Gain Observer Design for Nonlinear Two-Time-Scale Systems*," 2023 20th International Bhurban Conference on Applied Sciences and Technology (**IBCAST**), Bhurban, Murree, Pakistan, 2023, pp. 69-74, doi: 10.1109/IBCAST59916.2023.10712930.

Selected Research & Engineering Projects

- **Custom Drone & Flight Controller:** Designed custom flight controller hardware and firmware from scratch, implementing sensor fusion (IMU/Baro) and closed-loop PID attitude control algorithms for autonomous multirotor platforms.

- **Autonomous UGV with Robotic Arm:** Implemented ROS-based SLAM navigation (LiDAR/Vision) and 4-DOF manipulator kinematics for autonomous pick-and-place tasks.
- **Vision-Based Autonomous Target Tracking:** Deployed real-time object tracking algorithms (YOLO, OpenCV) integrated directly with flight control systems on an edge compute platform.
- **Dual Motor Control for RCWS:** Built closed-loop, high-precision dual-axis motor control system for coordinated pan-tilt actuation.

Technical Skills & Competencies

Category	Skills & Technologies
Robotics & Frameworks	ROS, ROS2, Gazebo, RViz, MoveIt, Autonomous Navigation, SLAM, Path Planning
Embedded & Programming	C, C++, Python, C#, STM32CubeIDE, Arduino IDE, DAVE IDE, VSCode, Embedded Linux
Control & Simulation	MATLAB, Simulink, Control System Toolbox, Hardware-in-the-Loop (HIL) Simulation
PCB Design & EDA	Altium Designer, KiCAD, EasyEDA, Simulation (NI Multisim, Proteus, LT-Spice)
Computer Vision & AI	OpenCV, YOLO, Feature Detection, Image Processing Pipeline Optimization

Honors, Awards & Presentations

- **Paper Presenter – iCoMET 2023:** Presented research paper *"Visual-Inertial State-Estimation Using Ground Station for UAV"* at the 4th International Conference on Computing, Mathematics and Engineering Technologies (Sukkur, Pakistan).
- **Paper Presenter – IBCAST 2023:** Presented research paper *"High-Gain Observer Design for Nonlinear Two-Time-Scale Systems"* at IBCAST-2023 (Islamabad, Pakistan).
- **Paper Presenter – ICAME 2023:** Presented research paper *"Output Feedback Control of Two-Time-Scale Permanent-Magnet DC Motor Using High-Gain Observers"* at ICAME 2023.
- **Gold Medalist:** HCRA English Quiz Competition (October 2018).

Certifications

- **Coursera:** Aerial Robotics | The Arduino Platform and C Programming | Introduction to Programming with MATLAB | Mastering Programming with MATLAB | Programming for Everybody (Python) | Python Data Structures
- **MathWorks:** MATLAB Fundamentals | Simulink Fundamentals

Academic References

Dr. Fahad Mumtaz Malik (Project Supervisor)

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Dr. Shehzad Amin Sheikh (Subject Teacher)

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